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Which Investors Drive Anomaly Returns and How?

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1 Big picture

1.1 Research question

The paper asks:

Which investor types are responsible for anomaly return variation, and through which trading motives do they affect prices?

The key difficulty is that common explanations for anomalies (risk-based, flow-based, heterogeneous-investor / mispricing) make different predictions about *who* should matter (retail vs institutions) and *how* they matter (fundamentals, sentiment/noise, flows, industry rotation), but it is empirically hard to put these explanations on a common scale.

1.2 Core contribution

The paper builds a unified empirical framework that:

1. estimates an **equity demand system** (nested logit with an industry-stock structure) using holdings data;
2. uses the implied **equilibrium price system** to run **counterfactual** updates of one driver at a time;
3. converts those counterfactual price changes into **anomaly-portfolio return components**;
4. performs a **time-series variance decomposition** of anomaly returns jointly by **trading motive** and **investor type**;
5. additionally links the results to a **cash-flow vs discount-rate (CF-DR) news decomposition**.

1.3 Headline findings (what to remember)

Across 47 anomaly portfolios grouped into 7 categories (value, momentum, profitability, investment, issuance, size, tangibility):

- **Fundamental-based trading (“demand for characteristics”)** is the **primary driver** of anomaly-return variance (about $\approx 40\%$ on average).
- **Non-fundamental trading (“latent demand”)** is the **second layer** (about $\approx 20\%$ on average), especially important where characteristics explain less (value and investment).
- **Institution-led channels (flows and industry rotation)** are **smaller on average**, though they matter in specific categories (flows in issuance/size; industry rotation in value/tangibility).
- **Small investors (a refined non-13F residual)** disproportionately transmit both the fundamental-based component and much of the latent-demand component.
- **Flow-induced trading is mostly institutional** (mutual funds and investment advisors).
- **CF news dominates DR news in explaining anomaly innovations**, and a large share of CF-related variation is tied to small-investor components.

2 Data objects and empirical targets

2.1 Investor groups (ownership construction)

The holdings universe is built from:

- quarterly holdings of all **13F filers** (Thomson Reuters S34), classified into: {investment advisors, mutual funds, banks, insurance companies, pension funds, other 13F};
- **non-13F blockholders** (stakes > 5%) identified from Forms 13D/13G (EDGAR), available consistently from 1999 onward;
- stock-level **short interest** used to define a **short-seller** category (to avoid attributing borrowed-and-sold shares to long-only owners).

After removing blockholders and shorts from the non-13F residual, the paper defines “**small investors**” as *non-13F, non-blockholder, long-only* ownership.

2.2 Anomaly portfolios

The paper studies 47 anomaly portfolios (long–short: top quintile minus bottom quintile of a sorting characteristic), grouped into 7 categories. Portfolio construction follows the anomalies in Freyberger et al. (2020) with modifications to align with quarterly holdings. (Exact anomaly lists are referenced in an external appendix, but the main paper reports category-level and anomaly-level decompositions.)

2.3 What is being decomposed

For each anomaly portfolio p , let $r_{p,t}$ be the portfolio *log* return at time t . The empirical goal is to decompose $\text{Var}(r_{p,t})$ into economically interpretable components indexed by:

1. **Trading motive k** : demand for characteristics, latent demand, flow-induced trading, industry effects (plus smaller supply/dividend terms);
2. **Investor type i** : small investors, banks, insurers, advisors, mutual funds, pensions, other 13F, short sellers, blockholders.

3 Model 1: a two-tier nested-logit demand system for equities

3.1 Portfolio weights: industry and within-industry choice

There are G industries (Fama–French 5 in the paper). Let $w_{i,t}(\gamma)$ be the share of investor i ’s equity wealth allocated to industry γ , and $w_{i,t}(n \mid \gamma)$ be the weight on stock n *conditional* on being in industry γ . The total portfolio weight on stock $n \in \gamma$ is

$$w_{i,t}(n) = w_{i,t}(n \mid \gamma) w_{i,t}(\gamma). \quad (1)$$

3.2 Inner nest: conditional logit within each industry

Within industry γ , the conditional logit demand is:

$$w_{i,t}(n \mid \gamma) = \frac{\exp \left\{ b_{i,\gamma,t} + \beta_{0,i,\gamma,t} p_t(n) + \beta_{x,i,\gamma,t}^\top x_t(n) + \epsilon_{i,t}(n \mid \gamma) \right\}}{1 + \sum_{m \in \mathcal{N}_{i,\gamma}} \exp \left\{ b_{i,\gamma,t} + \beta_{0,i,\gamma,t} p_t(m) + \beta_{x,i,\gamma,t}^\top x_t(m) + \epsilon_{i,t}(m \mid \gamma) \right\}}, \quad (2)$$

where:

- $p_t(n)$ is the (log) price of stock n (the paper treats price as market equity / log market cap in the demand system);
- $x_t(n)$ is a vector of observable stock characteristics (size, book, dividends/book, profitability, investment, beta, accruals, etc.);
- $\mathcal{N}_{i,\gamma}$ is the set of stocks that investor i potentially holds inside γ ;
- the “1” in the denominator is an *inside-industry outside option* for untracked/omitted securities.

Equivalently, the share-ratio form is:

$$\ln \frac{w_{i,t}(n \mid \gamma)}{w_{i,t}(0 \mid \gamma)} = b_{i,\gamma,t} + \beta_{0,i,\gamma,t} p_t(n) + \beta_{x,i,\gamma,t}^\top x_t(n) + \epsilon_{i,t}(n \mid \gamma). \quad (3)$$

Key modeling choice. The coefficients $\beta_{0,i,\gamma,t}$ and $\beta_{x,i,\gamma,t}$ are allowed to vary by industry γ , so the mapping from characteristics to demand is sector-specific.

3.3 Inner-nest price endogeneity and instrument

Prices are equilibrium objects and correlate with $\epsilon_{i,t}$. The paper uses a within-industry analogue of Koijen–Yogo’s counterfactual market-equity instrument: construct a counterfactual price $\hat{p}_t(n)$ that is orthogonal to investor i ’s idiosyncratic demand shock, and estimate (3) by nonlinear GMM each quarter.

(For notes: the object to remember is $\hat{p}_t(n)$, a counterfactual market equity based on other investors’ holdings.)

3.4 Inclusive value

For each investor i and industry γ , define the inclusive value:

$$\mathcal{I}_{i,t}(\gamma) = 1 + \sum_{m \in \mathcal{N}_{i,\gamma}} \exp \left\{ b_{i,\gamma,t} + \beta_{0,i,\gamma,t} p_t(m) + \beta_{x,i,\gamma,t}^\top x_t(m) + \epsilon_{i,t}(m \mid \gamma) \right\}. \quad (4)$$

This summarizes the attractiveness of the entire within-industry menu. A useful identity is $\mathcal{I}_{i,t}(\gamma) = 1/w_{i,t}(0 \mid \gamma)$: higher $\mathcal{I}$ means the tracked stocks are more attractive relative to the within-industry outside option.

3.5 Outer nest: allocating across industries

The outer nest allocates total equity wealth across industries using inclusive values:

$$w_{i,t}(\gamma) = \frac{\exp\{\alpha_{i,\gamma} + \eta_{i,t}(\gamma)\} \mathcal{I}_{i,t}(\gamma)^{\sigma_{i,\gamma}}}{\sum_{\gamma'} \exp\{\alpha_{i,\gamma'} + \eta_{i,t}(\gamma')\} \mathcal{I}_{i,t}(\gamma')^{\sigma_{i,\gamma'}}}, \quad \gamma = 1, \dots, G. \quad (5)$$

Here:

- $\eta_{i,t}(\gamma)$ is **industry latent demand**;
- $\sigma_{i,\gamma} \in (0, 1)$ governs cross-industry substitution intensity.

Special cases (intuition for σ):

- If $\sigma_{i,\gamma} = 1$ for all γ , the system collapses toward a simple logit with equal substitution across and within industries.
- If $\sigma_{i,\gamma} = 0$ for all γ , industries are segmented (no substitution across industries via $\mathcal{I}$).

Share-ratio form (relative to a reference industry, “Others”):

$$\ln \frac{w_{i,t}(\gamma)}{w_{i,t}(\text{Others})} = (\alpha_{i,\gamma} - \alpha_{i,\text{Others}}) + (\eta_{i,t}(\gamma) - \eta_{i,t}(\text{Others})) + \sigma_{i,\gamma} \ln \mathcal{I}_{i,t}(\gamma) - \sigma_{i,\text{Others}} \ln \mathcal{I}_{i,t}(\text{Others}). \quad (6)$$

3.6 Outer-nest instrument

Because $\mathcal{I}_{i,t}(\gamma)$ contains contemporaneous prices, it is instrumented using predicted inclusive values:

$$\hat{\mathcal{I}}_{i,t}(\gamma) = 1 + \sum_{m \in \mathcal{N}_{i,\gamma}} \exp \left\{ b_{i,\gamma,t} + \beta_{0,i,\gamma,t} \hat{p}_t(m) + \beta_{x,i,\gamma,t}^\top x_t(m) \right\}, \quad (7)$$

i.e., replace $p_t(m)$ with the counterfactual $\hat{p}_t(m)$ from the inner-nest instrument construction.

4 Model 2: the equilibrium price system and a counterfactual decomposition

4.1 The pricing map

Let:

- $\mathbf{p}_t$: $N \times 1$ vector of equilibrium log prices,
- $\mathbf{s}_t$: $N \times 1$ vector of log shares outstanding,
- $\mathbf{X}_t$: $N \times K$ matrix of characteristics,
- $\mathbf{A}_t$: vector of investors’ AUM,
- β_t : inner-nest coefficients on characteristics (stacked),
- ϵ_t : within-industry (stock-level) demand shocks,
- $\mathcal{J}_t \equiv (\boldsymbol{\eta}_t, \boldsymbol{\alpha}_t, \boldsymbol{\sigma}_t)$: outer-nest block (industry latent demand, intercepts, substitution).

The nested-logit equilibrium implies a pricing map:

$$\mathbf{p}_t = \mathbf{g}(\mathbf{s}_t, \mathbf{X}_t, \mathbf{A}_t, \beta_t, \epsilon_t; \mathcal{J}_t). \quad (8)$$

4.2 Returns

Quarterly log returns are:

$$\mathbf{r}_t = \mathbf{p}_t - \mathbf{p}_{t-1} + \mathbf{v}_t, \quad (9)$$

where $\mathbf{v}_t$ is the vector of log dividend yields.

4.3 Two-loop counterfactual sequence

To attribute $\mathbf{p}_t - \mathbf{p}_{t-1}$ to economically distinct drivers, the paper updates inputs sequentially:

- **Outer loop:** first update the entire outer block $\mathcal{J}$ (industry effects).
- **Inner loop:** holding $\mathcal{J}$ fixed, update in order: $\mathbf{s}$ (shares outstanding), $\mathbf{X}$ (characteristics), $\mathbf{A}$ (AUM), β (demand for characteristics), ϵ (latent demand).

After each update, solve for the new equilibrium price vector using a Newton solver adapted to the nested system.

4.4 Decomposition identity

Let $\Theta_{t-1} \equiv (\mathbf{s}_{t-1}, \mathbf{X}_{t-1}, \mathbf{A}_{t-1}, \beta_{t-1}, \epsilon_{t-1}; \mathcal{J}_{t-1})$. Then:

$$\mathbf{p}_t - \mathbf{p}_{t-1} = \Delta \mathbf{p}_t(\mathcal{J}) + \Delta \mathbf{p}_t(\mathbf{s}) + \Delta \mathbf{p}_t(\mathbf{X}) + \Delta \mathbf{p}_t(\mathbf{A}) + \Delta \mathbf{p}_t(\beta) + \Delta \mathbf{p}_t(\epsilon) + \psi_t, \quad (10)$$

where each component is defined as the difference between the equilibrium prices before and after updating that block (in the two-loop order), and ψ_t is a residual term capturing numerical error and higher-order effects.

4.5 Economic labels: “trading motives”

The paper labels the demand-side components as **trading motives**:

- $\Delta \mathbf{p}_t(\beta)$: **demand for stock characteristics** (fundamental-based trading),
- $\Delta \mathbf{p}_t(\epsilon)$: **latent demand** (non-fundamental-based trading),
- $\Delta \mathbf{p}_t(\mathbf{A})$: **flow-induced trading** (scale/AUM pressure),
- $\Delta \mathbf{p}_t(\mathcal{J})$: **industry effects** (outer-nest reallocation; includes η, α, σ).

The remaining terms (shares outstanding, characteristics, dividends) are grouped as **other effects**.

5 Model 3: decomposition by investor type

5.1 Group-switch counterfactuals

To isolate investor-group contributions, the paper performs *one-group-at-a-time* updates: switch only group g ’s component of a motive from $t-1$ to t , holding all other groups at $t-1$, and re-solve equilibrium.

Schematically, for a generic motive block $(\cdot) \in \{\mathcal{J}, \mathbf{A}, \beta, \epsilon\}$,

$$\Delta \mathbf{p}_t((\cdot)_g) = \mathbf{g}\left(\text{state with } (\cdot)_g \text{ at } t, (\cdot)_{-g} \text{ at } t-1\right) - \mathbf{g}\left(\text{all at } t-1\right). \quad (11)$$

Interpretation. Each group-level effect includes both: (i) the direct effect of the group-specific demand shift, and (ii) the indirect effect through equilibrium price changes that alter market-wide elasticities.

6 Mapping to anomaly portfolios and variance decomposition

6.1 Portfolio return induced by a (motive, investor) component

For anomaly portfolio p , let $w_{n,t-1}^{(p)}$ be the portfolio weight on stock n at $t-1$. Given a component-induced stock-level log price change $\Delta p_{n,t}(k, i)$, the induced *simple* portfolio return is

$$R_{p,t}^{(k,i)} = \sum_{n=1}^N w_{n,t-1}^{(p)} (\exp\{\Delta p_{n,t}(k, i)\} - 1), \quad (12)$$

and the induced *log* return is $r_{p,t}^{(k,i)} = \log(1 + R_{p,t}^{(k,i)})$.

The total portfolio log return decomposes as:

$$r_{p,t} = \sum_k \sum_i r_{p,t}^{(k,i)} + u_{p,t}, \quad (13)$$

where $u_{p,t}$ collects residual terms (including the model-price residual and log mapping differences).

6.2 Time-series variance decomposition

The variance identity used in the paper is:

$$\text{Var}(r_{p,t}) = \sum_k \sum_i \text{Cov}(r_{p,t}^{(k,i)}, r_{p,t}) + \text{Cov}(u_{p,t}, r_{p,t}). \quad (14)$$

To estimate shares, for each (k, i) run:

$$r_{p,t}^{(k,i)} = a_{k,i} + b_{k,i} r_{p,t} + z_{k,i,t}, \quad b_{k,i} = \frac{\text{Cov}(r_{p,t}^{(k,i)}, r_{p,t})}{\text{Var}(r_{p,t})}. \quad (15)$$

Thus $b_{k,i}$ is interpreted as the **share of variance** attributable to the (k, i) component. Shares sum to one across all components plus the residual share.

Aggregating across investors yields motive-level shares:

$$b_k = \sum_i b_{k,i}. \quad (16)$$

7 Cash-flow vs discount-rate news (CF–DR) integration

7.1 CF–DR decomposition for log return innovations

Using the Campbell (1991) present-value framework, innovations in log returns satisfy approximately:

$$\tilde{r}_{n,t} \equiv r_{n,t} - \mathbb{E}_t[r_{n,t}] \approx cf_{n,t} - dr_{n,t}, \quad (17)$$

where $cf_{n,t}$ is cash-flow news and $dr_{n,t}$ is discount-rate news. The paper estimates the required expectations via a VAR and constructs stock-level CF/DR news (with fixed effects to absorb persistent heterogeneity).

7.2 Embedding investor–motive components

Replace $r_{n,t}$ with the component-induced $r_{n,t}^{(k,i)}$, aggregate to portfolios, and obtain $\tilde{r}_{p,t}^{(k,i)}$ along with portfolio-level $cf_{p,t}$ and $dr_{p,t}$. Then:

$$\text{Var}(\tilde{r}_{p,t}) = \sum_k \sum_i \text{Cov}(\tilde{r}_{p,t}^{(k,i)}, cf_{p,t}) + \sum_k \sum_i \text{Cov}(\tilde{r}_{p,t}^{(k,i)}, -dr_{p,t}) + \text{Cov}(u_{p,t}, \tilde{r}_{p,t}), \quad (18)$$

so each investor–motive component can be associated with CF-driven vs DR-driven variance contributions.

8 Read the empirical results: figures and tables

8.1 Figure 1: variance decomposition by trading motive

What it shows. For each anomaly category, Figure 1 reports the share of return variance explained by: demand for characteristics, latent demand, flow-induced trading, industry effects, other effects, and residual.

Main interpretation.

- The dominant bar segment across most categories is **demand for characteristics**, especially strong for momentum and still large for profitability, size, and issuance.
- **Latent demand** rises precisely where characteristic-demand weakens (notably value and investment), consistent with an additional non-fundamental layer.
- **Flows** are small on average and become material mainly for **issuance** and **size**.
- **Industry effects** are moderate and peak in categories where cross-industry rotation is plausibly important (value and tangibility).

8.2 Table 1: variance decomposition of anomaly returns (anomaly-by-anomaly)

What it is. Table 1 is the anomaly-level version of Figure 1: it reports, for each of the 47 anomalies, the variance shares for: demand for characteristics, flow-induced trading, latent demand, and industry effects, plus category means and overall percentiles.

How I interpret it.

- The **overall mean** across all anomalies is approximately: $\approx 40\%$ demand-for-characteristics, $\approx 19\%$ latent demand, $\approx 13\%$ industry effects, and only $\approx 3\%$ flow-induced trading.
- The **interquartile ranges** show hierarchy and dispersion: characteristic demand is both large and variable across anomalies, latent demand is the next most variable component, industry effects are steadier and smaller, and flows are tightly clustered near zero with only a few outliers.
- Negative shares for some components mean the component *offsets* total variation (e.g., latent demand for size is negative in the category mean).

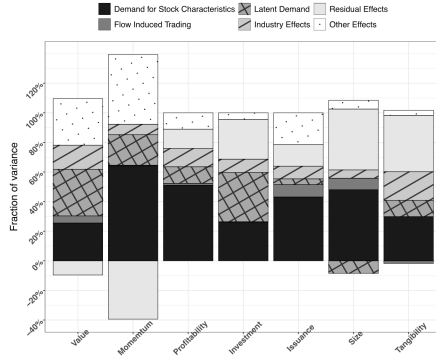


Fig. 1. Variance Decomposition by Trading Motives. Each bar represents a specific anomaly group. Appendix Table A.1 lists the anomalies included in each

Table 1
Variance Decomposition of Anomaly Returns
This table reports the variance decomposition of the 47 anomaly portfolios, grouped into 7 categories. Appendix Table A.1 lists the anomalies included in each group. The estimates represent the proportions of total anomaly return variance attributed to demand for stock characteristics, flow-induced trading, latent demand, and industry effects. Detailed information can be found in Section 2. We estimate standard errors for the cross-anomaly means using a seemingly unrelated regressions (SUR) framework with Newey-West (one-lag) correction to account for within-anomaly serial correlation. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Anomalies	Demand for stock characteristics	Flow-induced trading	Latent demand	Industry effects
A. Value				
Total assets over market equity	12%	4%	15%	19%
Book equity over market equity	17%	4%	51%	13%
Industry-adjusted book-to-market ratio	24%	6%	72%	4%
Cash and short-term investments over total assets	19%	6%	34%	18%
Cash flow to total liabilities	32%	6%	4%	12%
Log change in the shares outstanding	53%	7%	15%	15%
Debt over market equity	11%	6%	20%	17%
Income over market equity	12%	11%	48%	15%
Cash flow to book equity	70%	2%	-2%	14%
Net payout ratio	31%	7%	34%	26%
Payout ratio	32%	8%	31%	24%
Tobin's Q	20%	3%	50%	11%
Sales over market equity	17%	3%	46%	18%
Sales growth	7%	1%	24%	22%
Mean	26%***	5%***	32%***	16%***
B. Momentum				
12-month lagged returns, held for 12 months	53%	1%	28%	6%
6-month lagged returns, held for 6 months	65%	0%	19%	9%
12-month lagged returns, held for 3 months	75%	0%	15%	6%
Mean	64%***	0%***	21%***	7%***
C. Profitability				
Sales over lagged net operating assets	44%	0%	32%	24%
Sales over lagged total assets	24%	0%	12%	1%
Differences in growth of gross margin and sales	3%	2%	72%	9%
Earnings per share	50%	6%	-3%	12%
Pre-tax income over sales	70%	3%	-7%	5%
Price-to-cost margin	28%	0%	-1%	22%
Profit margin	42%	-1%	38%	6%
Industry-adjusted profit margin	14%	6%	11%	36%
Gross profitability over book equity	116%	0%	-3%	12%
Returns on net operating assets	81%	0%	-23%	16%
Returns on assets	6%	-6%	0%	1%
Returns on capital	-3%	8%	49%	25%
Returns on equity	79%	1%	1%	0%
Returns on invested capital	77%	-8%	-3%	3%
Sales over cash and short-term investments	25%	3%	38%	17%
Sales over total assets	41%	-2%	13%	5%
Quality score from (Jensen et al., 2019)	95%	-4%	-22%	5%
Mean	51%***	1%	11%	12%***
D. Investment				
% Change in total assets	29%	4%	3%	31%
% Change in book equity	5%	-3%	66%	-7%
Change in P/BK over lagged total assets	21%	2%	66%	12%
Change in inventories over average total assets	21%	3%	42%	2%
Net operating assets over lagged total assets	53%	-6%	-11%	7%
Mean	30%***	0%	33%***	9%
E. Insurance				
1-year net-share insurance	50%	8%	22%	6%
5-year composite-share insurance	30%	9%	-14%	11%
Mean	43%***	8%***	4%	9%***
F. Size				
Total assets	42%	12%	-26%	6%
Market equity	46%	2%	-17%	6%
Industry-adjusted market equity	50%	8%	18%	6%
Mean	48%***	8%***	-8%	6%***
G. Tangibility				
Absolute value of operating accruals	30%	-2%	21%	28%

8.3 Figure 2: who drives each motive? (investor $\times$ motive)

Structure. Figure 2 has four panels, one per main motive (characteristics, latent, flows, industry). Each bar is a category; bar height is the total motive share (from Figure 1); stack colors split that motive across investor groups.

Key message. Different motives are transmitted by different investors:

- **Characteristics demand:** mostly **small investors**.
- **Latent demand:** still tilted to **small investors** in categories where it is large, with some institutional participation.
- **Flow-induced trading:** mostly **mutual funds and investment advisors**.
- **Industry effects:** predominantly **institutions** (mutual funds, advisors, banks), with a smaller small-investor slice.

8.4 Table 2: demand for characteristics by investor

What it is. Table 2 reports the variance shares of the *characteristics-demand* channel for each investor group.

Interpretation.

- Across all anomalies, **small investors** have by far the largest mean share (roughly $\approx 27\%$).
- Banks and mutual funds are positive but much smaller (each around $\approx 5\%$ on average), and several groups are near zero.
- The dispersion (25th–75th percentiles) is wide for small investors and tight for institutions, reinforcing that cross-anomaly variation in the fundamentals channel is largely a small-investor phenomenon.

Table 2

The Effects of Demand for Stock Characteristics by Investor.

This table reports the decomposition of the effects of demand for stock characteristics by investor across 47 anomaly portfolios, grouped into 7 categories. The estimates represent the proportions of total anomaly return variance attributed to demand of each investor type. Detailed information can be found in Section 2. We estimate standard errors for the cross-anomaly means using a seemingly unrelated regression (SUR) framework with Newey-West (one-lag) correction to account for within-anomaly serial correlation. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Anomalies	Small investors	Banks	Insurance companies	Investment advisors	Mutual funds	Pension funds	Other investors	Short sellers	Block holders
A. Value									
Total assets over market equity	7%	2%	1%	1%	2%	0%	0%	-1%	0%
Book equity over market equity	-3%	8%	1%	6%	7%	0%	0%	-2%	0%
Industry-adjusted book-to-market ratio	-10%	12%	2%	9%	11%	2%	0%	-2%	0%
Cash and short-term investments over total assets	-2%	2%	2%	10%	6%	1%	0%	0%	0%
Cash flow to total liabilities	28%	3%	1%	-3%	2%	0%	1%	0%	0%
Log change in the share outstanding	40%	6%	1%	2%	3%	1%	0%	-1%	0%
Debt over market equity	7%	2%	1%	2%	1%	0%	0%	-1%	0%
Income over market equity	5%	3%	1%	5%	0%	0%	0%	-1%	0%
Cash flow to book equity	56%	8%	1%	-2%	5%	1%	1%	0%	0%
Net payout ratio	25%	4%	1%	1%	0%	1%	0%	0%	0%
Payout ratio	25%	4%	1%	1%	0%	1%	0%	0%	0%
Tobin's Q	-4%	8%	2%	9%	6%	1%	1%	-3%	0%
Sales over market equity	6%	0%	0%	3%	-1%	0%	1%	0%	0%
Sales growth	0%	2%	1%	2%	1%	0%	0%	0%	0%
Mean	13%***	5%***	1%***	3%***	1%***	0%***	-1%***	0%	0%
B. Momentum									
12-month lagged returns, held for 12 months	38%	4%	1%	3%	3%	1%	0%	0%	0%
6-month lagged returns, held for 6 months	55%	1%	1%	6%	1%	0%	0%	0%	0%
12-month lagged returns, held for 3 months	63%	2%	1%	2%	4%	1%	0%	0%	0%
Mean	52%***	2%***	1%***	2%***	3%***	1%***	0%	0%***	0%***
C. Profitability									
Sales over lagged net operating assets	25%	4%	2%	5%	7%	0%	0%	0%	0%
Sales over lagged total assets	11%	4%	2%	2%	5%	1%	0%	-1%	0%
Difference in growth of gross margin and sales	62%	4%	0%	6%	1%	0%	0%	0%	0%
Earnings per share	37%	4%	1%	0%	6%	0%	1%	0%	0%
Pre-tax income over sales	42%	0%	0%	0%	0%	0%	0%	0%	0%
Price-to-cost margin	27%	2%	1%	-3%	1%	1%	0%	-1%	0%
Profit margin	42%	3%	0%	-2%	-3%	-1%	0%	1%	0%
Industry-adjusted profit margin	9%	3%	1%	3%	1%	0%	0%	0%	0%
Gross profitability over book equity	56%	19%	2%	7%	30%	1%	0%	0%	0%
Return on net operating assets	47%	8%	3%	11%	12%	1%	0%	0%	0%
Return on assets	66%	5%	2%	0%	7%	1%	1%	1%	0%
Return on capital	-28%	7%	2%	12%	4%	1%	0%	-1%	0%
Return on equity	62%	5%	1%	-1%	10%	-1%	1%	1%	0%
Return on invested capital	59%	5%	1%	0%	12%	0%	1%	0%	0%
Sales over cash and short-term investments	20%	-1%	0%	2%	4%	1%	0%	0%	0%
Sales over total assets	23%	3%	0%	3%	11%	0%	0%	-1%	0%
Quality score from Kaneva et al., 2019	74%	4%	2%	-1%	11%	1%	1%	1%	0%
Mean	32%***	5%***	1%***	2%	7%***	0%***	0%***	0%	0%
D. Investment									
% Change in total assets	5%	8%	2%	7%	5%	1%	1%	0%	0%
% Change in book equity	11%	1%	0%	2%	4%	1%	0%	0%	0%
Change in P&ID over lagged total assets	16%	1%	3%	0%	-4%	0%	0%	0%	0%
Change in inventories over average total assets	1%	7%	3%	3%	7%	2%	-1%	-1%	0%
Net operating assets over lagged total assets	27%	7%	1%	8%	9%	1%	0%	0%	0%
Mean	12%***	5%***	2%***	4%***	3%	1%***	0%	0%***	0%
E. Insurance									
1-year net share issuance	39%	5%	1%	2%	2%	1%	0%	-1%	0%
5-year composite-share issuance	52%	5%	1%	-21%	-1%	0%	0%	0%	0%
Mean	46%***	5%***	1%***	-10%	1%	1%	0%	-1%	0%
F. Size									
Total assets	39%	4%	-1%	-1%	1%	0%	0%	0%	0%
Market equity	39%	1%	0%	4%	3%	0%	0%	0%	0%
Industry-adjusted market equity	51%	5%	0%	0%	3%	0%	1%	-3%	0%
Mean	43%***	3%***	0%	1%	2%***	0%	0%	-1%	0%
G. Tangibility									
Absolute value of operating accruals	33%	1%	0%	-2%	-2%	0%	0%	0%	0%
Operating leverage	3%	3%	-1%	7%	17%	0%	0%	-1%	0%

Table 3

The Effects of Latent Demand by Investor.

This table reports the decomposition of the latent demand by investor across 47 anomaly portfolios, grouped into 7 categories. The estimates represent the proportions of total anomaly return variance attributed to demand of each investor type. Detailed information can be found in Section 2. We estimate standard errors for the cross-anomaly means using a seemingly unrelated regression (SUR) framework with Newey-West (one-lag) correction to account for within-anomaly serial correlation. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Anomalies	Small investors	Banks	Insurance companies	Investment advisors	Mutual funds	Pension funds	Other investors	Short sellers	Block holders
A. Value									
Total assets over market equity	3%	7%	1%	0%	2%	1%	0%	2%	0%
Book equity over market equity	20%	8%	4%	3%	10%	2%	0%	4%	0%
Industry-adjusted book-to-market ratio	52%	1%	4%	2%	9%	1%	1%	1%	0%
Cash and short-term investments over total assets	3%	11%	3%	6%	7%	1%	1%	2%	0%
Cash flow to total liabilities	17%	-8%	0%	-3%	-1%	0%	-1%	0%	0%
Log change in the share outstanding	6%	4%	4%	-14%	12%	-1%	-2%	6%	0%
Debt over market equity	0%	7%	2%	4%	4%	1%	0%	2%	0%
Income over market equity	5%	13%	3%	8%	16%	1%	0%	3%	0%
Cash flow to book equity	1%	-5%	5%	-1%	-1%	0%	-2%	1%	0%
Net payout ratio	13%	6%	2%	-2%	12%	0%	0%	2%	0%
Payout ratio	8%	4%	3%	2%	14%	0%	-1%	1%	0%
Tobin's Q	20%	7%	3%	5%	9%	1%	1%	4%	0%
Sales over market equity	12%	6%	3%	10%	11%	0%	0%	4%	0%
Sales growth	1%	4%	7%	-2%	8%	1%	0%	2%	0%
Mean	12%***	5%***	3%***	1%	8%***	1%***	0%	2%***	0%***
B. Momentum									
12-month lagged returns, held for 12 months	16%	0%	1%	3%	8%	0%	1%	0%	0%
6-month lagged returns, held for 6 months	19%	-7%	2%	2%	2%	0%	1%	-1%	0%
12-month lagged returns, held for 3 months	12%	-3%	1%	-2%	6%	0%	1%	-1%	0%
Mean	16%***	-3%	1%***	1%	5%*	0%	1%***	-1%***	0%***
C. Profitability									
Sales over lagged net operating assets	12%	8%	-3%	3%	12%	1%	0%	0%	0%
Sales over lagged total assets	14%	6%	-2%	-3%	-1%	0%	0%	-1%	0%
Difference in growth of gross margin and sales	49%	15%	2%	1%	1%	0%	1%	2%	0%
Earnings per share	1%	-2%	3%	-4%	-2%	0%	-1%	2%	0%
Pre-tax income over sales	1%	-11%	6%	-3%	0%	0%	-2%	2%	0%
Price-to-cost margin	-43%	27%	-2%	0%	2%	1%	1%	4%	0%
Profit margin	7%	22%	6%	-1%	2%	0%	0%	1%	0%
Industry-adjusted profit margin	-2%	6%	1%	0%	5%	1%	0%	0%	0%
Gross profitability over book equity	-2%	1%	3%	0%	-9%	1%	1%	0%	0%
Return on net operating assets	-7%	-18%	-7%	-1%	5%	1%	0%	0%	0%
Return on assets	0%	-8%	2%	-5%	2%	0%	-1%	2%	0%
Return on capital	16%	6%	3%	0%	12%	2%	1%	3%	0%
Return on equity	7%	-11%	4%	-3%	4%	0%	-2%	3%	0%
Return on invested capital	4%	-2%	2%	-6%	-2%	0%	0%	0%	0%
Sales over cash and short-term investments	5%	15%	0%	10%	8%	0%	0%	0%	0%
Sales over total assets	11%	6%	-2%	2%	0%	0%	0%	-1%	0%
Quality score from Kaneva et al., 2019	0%	-7%	2%	-14%	-4%	0%	-1%	2%	0%
Mean	4%	3%	1%	0%	2%	1%***	0%	1%***	0%***
D. Investment									
% Change in total assets	-6%	-5%	6%	-1%	8%	0%	-1%	3%	0%
% Change in book equity	20%	9%	2%	-5%	30%	1%	0%	4%	0%
Change in P&ID over lagged total assets	52%	-2%	3%	10%	4%	-1%	-2%	2%	0%
Change in inventories over average total assets	6%	16%	-2%	7%	9%	1%	1%	4%	0%
Net operating assets over lagged total assets	-2%	-6%	0%	2%	0%	0%	-1%	0%	0%
Mean	13%*	2%	0%	2%	11%***	0%	-1%	3%***	0%*
E. Insurance									
1-year net share issuance	4%	7%	3%	-15%	16%	0%	-3%	7%	0%
5-year composite-share issuance	13%	-12%	-1%	-11%	-2%	-1%	0%	0%	0%

(continued on next page)

8.5 Table 3: latent demand by investor

What it is. Table 3 reports the variance shares of the *latent-demand* channel for each investor group.

Interpretation.

- Latent demand is most important (and small-investor driven) in **value** and **investment**, consistent with a non-fundamental/sentiment-like layer where characteristics explain less.
- The size category is unusual: the mean latent-demand contribution for small investors is negative, while banks are positive, implying an offsetting pattern across investors.
- Mutual funds appear as a recurring secondary contributor to latent demand in several categories, but typically below small investors when latent demand is large.

8.6 Table 4: flow-induced trading by investor

What it is. Table 4 reports the variance shares of the *AUM/flow* channel for each investor group.

Interpretation.

- The average flow-induced trading share is **small** and concentrated in **issuance** and **size**.
- The dominant contributors are **mutual funds** and **investment advisors** (institutional delegated management).
- Small-investor flow effects are near zero or slightly negative, consistent with the idea that “flow” is an institutional margin in this setting.

Table 4

The Effects of Flow-Induced Trading By Investor.

This table reports the decomposition of the effects of flow induced trading by investor across 47 anomaly portfolio, grouped into 7 categories. The estimates represent the proportions of total anomaly returns attributed to flow-induced trading of each investor type. Detailed information can be found in Section 2. We estimate standard errors for the cross-anomaly means using a seemingly unrelated regressions (SUR) framework with Newey-West (one-lag) corrections to account for within-anomaly serial correlation. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Anomaly	Small investors	Banks	Insurance companies	Investment advisors	Mutual funds	Pension funds	Other investors	Short sellers	Block holders
A. Value									
Total assets over market equity	4%	-1%	0%	0%	0%	0%	0%	0%	0%
Book equity over market equity	-2%	0%	0%	2%	4%	0%	0%	0%	0%
Industry-adjusted book-to-market ratio	0%	-1%	0%	1%	3%	0%	0%	0%	0%
Cash and short-term investments over total assets	-1%	3%	1%	0%	3%	0%	0%	-1%	0%
Cash flow to total liabilities	2%	-1%	0%	0%	-1%	0%	0%	1%	0%
Log change in the shares outstanding	-4%	-1%	0%	5%	5%	0%	0%	1%	0%
Debt over market equity	7%	-1%	0%	0%	0%	0%	0%	0%	0%
Income over market equity	4%	1%	0%	1%	4%	0%	0%	0%	0%
Cash flow to book equity	-11%	-4%	0%	8%	9%	-1%	0%	1%	0%
Net payout ratio	-2%	-1%	0%	4%	6%	0%	0%	0%	0%
Payout ratio	3%	-2%	0%	3%	4%	0%	0%	0%	0%
Tabler's Q	-3%	-1%	0%	1%	3%	0%	0%	1%	0%
Sales over market equity	-5%	0%	0%	1%	3%	0%	0%	2%	0%
Sales growth	-3%	0%	1%	1%	3%	0%	0%	0%	0%
Mean	-1%	-1%	0%	2%	4%	0%	0%	0%	0%
B. Momentum									
12-month lagged returns, held for 12 months	-3%	0%	0%	1%	1%	0%	0%	1%	0%
6-month lagged returns, held for 6 months	-2%	-1%	0%	2%	1%	0%	0%	1%	0%
12-month lagged returns, held for 3 months	-2%	-1%	0%	2%	0%	0%	0%	1%	0%
Mean	-2%	-1%	0%	2%	1%	0%	0%	1%	0%
C. Profitability									
Sales over lagged net operating assets	3%	0%	1%	3%	3%	0%	0%	-1%	0%
Sales over lagged total assets	3%	-1%	0%	0%	1%	0%	0%	0%	0%
Differences in growth of gross margin and sales	2%	0%	0%	0%	-1%	0%	0%	0%	0%
Earnings per share	-5%	-2%	0%	0%	8%	0%	0%	1%	1%
Pre-tax income over sales	-11%	-3%	0%	7%	7%	-1%	0%	3%	0%
Price-to-cost margin	-2%	-1%	0%	1%	1%	0%	0%	1%	0%
Profit margin	-8%	-3%	0%	3%	5%	-1%	0%	1%	0%
Industry-adjusted profit margin	1%	1%	0%	0%	3%	0%	0%	0%	0%
Gross profitability over book equity	3%	-3%	-1%	1%	-1%	0%	0%	0%	0%
Returns on net operating assets	2%	-3%	0%	1%	-1%	0%	0%	0%	0%
Returns on assets	-8%	-3%	0%	3%	-1%	0%	0%	0%	0%
Returns on capital	4%	2%	0%	0%	2%	0%	0%	0%	0%
Returns on equity	-10%	-4%	-1%	6%	7%	-1%	0%	2%	0%
Returns on invested capital	-2%	-5%	0%	1%	-2%	-1%	0%	1%	0%
Sales over cash and short-term investments	3%	1%	0%	0%	0%	0%	-1%	0%	0%
Sales over total assets	0%	-1%	0%	0%	-1%	0%	0%	0%	0%
Quality score from (Jensen et al., 2019)	-5%	-5%	0%	4%	0%	-1%	0%	3%	0%
Mean	-2%	-2%	0%	2%	2%	0%	0%	1%	0%
D. Investment									
% Change in total assets	-1%	-1%	0%	1%	3%	0%	0%	0%	0%
% Change in book equity	-3%	-1%	0%	-1%	2%	0%	0%	0%	0%
Change in FPRR over lagged total assets	-5%	0%	1%	3%	3%	0%	0%	0%	0%
Change in investments over average total assets	-3%	0%	0%	2%	3%	0%	0%	0%	0%
Net operating assets over lagged total assets	-3%	-5%	0%	0%	2%	-1%	0%	0%	0%
Mean	-3%	-1%	0%	1%	3%	0%	0%	0%	0%
E. Insurance									
1-year net-share insurance	-1%	-1%	0%	5%	4%	0%	0%	1%	0%
5-year composite-share insurance	-7%	-1%	1%	7%	10%	0%	0%	1%	0%
Mean	-4%	-1%	0%	6%	7%	0%	0%	1%	0%
F. Size									
Total assets	2%	-7%	-1%	8%	9%	-1%	0%	1%	1%
Market equity	11%	-7%	-1%	0%	0%	-2%	0%	1%	0%
Industry-adjusted market equity	1%	-7%	0%	7%	-1%	0%	0%	0%	0%
Mean	5%	-7%	-1%	4%	5%	-1%	0%	2%	0%
G. Tangibility									
Absolute value of operating accruals	-3%	-2%	-1%	1%	1%	0%	0%	1%	0%

Table 5

The Effects of Industry Demand By Investor.

This table reports the decomposition of the effects of industry demand by investor across 47 anomaly portfolio, grouped into 7 categories. The estimates represent the proportions of total anomaly return variance attributed to demand of each investor type. Detailed information can be found in Section 2. We estimate standard errors for the cross-anomaly means using a seemingly unrelated regressions (SUR) framework with Newey-West (one-lag) corrections to account for within-anomaly serial correlation. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Anomaly	Small investors	Banks	Insurance companies	Investment advisors	Mutual funds	Pension funds	Other investors	Short sellers	Block holders
A. Value									
Total assets over market equity	5%	4%	0%	2%	6%	1%	0%	0%	0%
Book equity over market equity	6%	3%	0%	0%	3%	0%	0%	0%	0%
Industry-adjusted book-to-market ratio	1%	1%	0%	0%	1%	0%	0%	0%	0%
Cash and short-term investments over total assets	8%	4%	0%	2%	3%	0%	0%	0%	0%
Cash flow to total liabilities	1%	4%	0%	2%	5%	0%	0%	0%	0%
Log change in the shares outstanding	4%	3%	1%	2%	6%	0%	0%	0%	0%
Debt over market equity	3%	4%	0%	2%	6%	1%	0%	0%	0%
Income over market equity	6%	3%	0%	1%	3%	0%	0%	0%	0%
Cash flow to book equity	6%	1%	0%	3%	3%	0%	0%	0%	0%
Net payout ratio	11%	4%	0%	3%	6%	1%	0%	0%	0%
Payout ratio	11%	3%	0%	3%	6%	1%	0%	0%	0%
Tabler's Q	3%	2%	0%	-1%	3%	0%	0%	0%	0%
Sales over market equity	7%	2%	1%	2%	6%	0%	0%	0%	0%
Sales growth	11%	4%	0%	2%	4%	0%	0%	0%	0%
Mean	6%	3%	0%	2%	4%	0%	0%	0%	0%
B. Momentum									
12-month lagged returns, held for 12 months	1%	1%	0%	1%	2%	0%	0%	0%	0%
6-month lagged returns, held for 6 months	4%	1%	0%	0%	2%	0%	0%	0%	0%
12-month lagged returns, held for 3 months	2%	1%	0%	1%	2%	0%	0%	0%	0%
Mean	2%	1%	0%	1%	2%	0%	0%	0%	0%
C. Profitability									
Sales over lagged net operating assets	14%	2%	0%	2%	4%	0%	0%	0%	0%
Sales over lagged total assets	-1%	0%	0%	3%	3%	0%	0%	0%	0%
Differences in growth of gross margin and sales	0%	3%	0%	3%	3%	0%	0%	0%	0%
Earnings per share	3%	2%	0%	3%	4%	0%	0%	0%	0%
Pre-tax income over sales	-3%	0%	0%	4%	3%	0%	0%	0%	0%
Price-to-cost margin	7%	4%	1%	3%	7%	0%	0%	0%	0%
Profit margin	-4%	3%	0%	3%	3%	0%	0%	0%	0%
Industry-adjusted profit margin	18%	5%	1%	2%	10%	1%	0%	0%	0%
Gross profitability over book equity	-1%	4%	0%	4%	4%	0%	0%	1%	0%
Returns on net operating assets	8%	2%	0%	2%	6%	0%	0%	0%	0%
Returns on assets	-5%	1%	0%	2%	6%	0%	0%	0%	0%
Returns on capital	13%	5%	0%	1%	4%	1%	0%	0%	0%
Returns on equity	-6%	1%	0%	2%	3%	0%	0%	0%	0%
Returns on invested capital	-2%	1%	0%	2%	3%	0%	0%	0%	0%
Sales over cash and short-term investments	7%	4%	1%	2%	3%	0%	0%	0%	0%
Sales over total assets	-3%	3%	0%	3%	3%	0%	0%	0%	0%
Quality score from (Jensen et al., 2019)	-2%	1%	0%	3%	2%	0%	0%	0%	0%
Mean	1%	2%	0%	2%	3%	0%	0%	0%	0%
D. Investment									
% Change in total assets	20%	3%	1%	1%	6%	0%	0%	0%	0%
% Change in book equity	-8%	1%	0%	1%	0%	0%	0%	0%	0%
Change in FPRR over lagged total assets	2%	4%	0%	1%	0%	0%	0%	0%	0%
Change in investments over average total assets	-4%	2%	0%	0%	3%	0%	0%	0%	0%
Net operating assets over lagged total assets	-2%	3%	0%	2%	3%	0%	0%	0%	0%
Mean	2%	3%	0%	1%	3%	0%	0%	0%	0%
E. Insurance									
1-year net-share insurance	-1%	1%	0%	1%	4%	0%	0%	0%	0%
5-year composite-share insurance	5%	1%	0%	2%	3%	0%	0%	0%	0%

8.7 Table 5: industry demand by investor

What it is. Table 5 decomposes the *industry (outer-nest)* channel by investor group.

Interpretation.

- Industry effects are **institution-led**: mutual funds, advisors, and banks have positive mean shares.
- Small investors contribute meaningfully in some categories (e.g., where retail rotates across sectors), but institutions dominate on average.
- Dispersion is smaller than in the characteristics-demand channel, suggesting a more stable, lower-variance industry-rotation mechanism.

8.8 Figure 3: CF vs DR news by investor type and trading motive

What it does. Figure 3 splits the main investor-motive effects into **cash-flow news** vs **discount-rate news** components.

Intuition.

- If anomaly variation is mostly CF-driven, it points toward mechanisms tied to (perceived or actual) profitability/cash-flow dynamics.
- If mostly DR-driven, it points toward time-varying risk premia / discounting.

Interpretation of the pattern reported in the paper. CF news dominates DR news across anomaly categories, and a large share of CF-driven variation is attributed to **small investors**, especially through the fundamental-based (characteristics-demand) channel. The latent-demand layer is also primarily CF-related rather than purely DR-related.

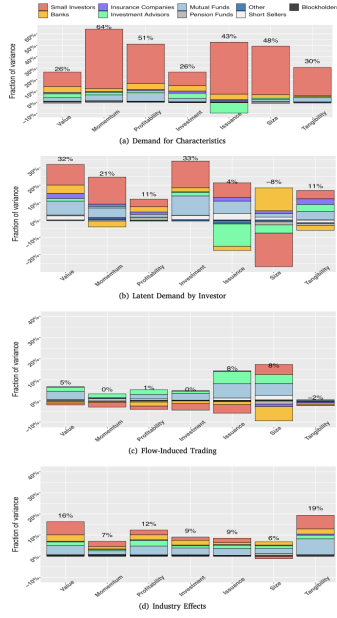


Fig. 2. Variance Decomposition by Investor and Trading Motive Panel (a) shows the effects of demand for stock characteristics by investor group. Panel (b) shows the effects of latent demand by investor group. Each bar represents a specific anomaly group. Appendix Table A.1 lists the anomalies included in each group. We combine individual anomalies into groups using the method described in Section 2.3.2. The bars display the fractions of total anomaly return variance explained by the effects of each investor type. The numbers above each bar indicate the total effects of each demand component from Fig. 1.

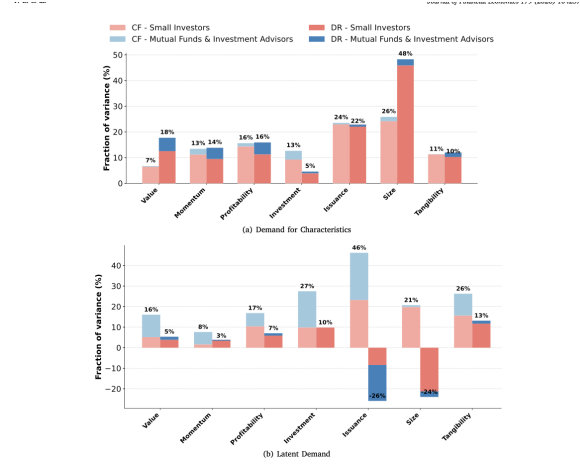


Fig. 3. Cash-Flow vs. Discount-Rate News by Investor Type and Trading Motive This figure reports a two-step variance decomposition of anomaly returns. Step 1 allocates total anomaly returns to trading motive and investor types. Step 2 splits the innovations in those motive-investor effects into cash-flow (CF) and discount-rate (DR) news. Panel (a) shows the fundamental-based component (demand for characteristics); Panel (b) shows the non-fundamental component (latent demand). Within each panel, we compare small investors to major institutions (mutual funds and investment advisors). Bars report the share of variance (by anomaly category) attributable to each motive-investor-news component; numbers above bars give the total motive effect for that investor type. Anomaly categories follow Section 2.3.2.

9 How the paper changes the mental map of anomaly theories

Within this framework, three theory families map to different channels:

- **Flow-based theories:** predict large flow-induced trading effects and institutional dominance. The evidence finds flows exist but are *secondary* and concentrated in a few categories.
- **Risk-based theories:** can be consistent with large characteristics-demand shares (if characteristics proxy for priced cash-flow/discount-rate risk), but the strong **small-investor concentration** is harder to reconcile with representative-investor versions.
- **Heterogeneous-investor / mispricing theories:** naturally predict (i) a large role for less sophisticated investors acting on fundamentals (possibly imperfectly), (ii) a meaningful non-fundamental layer, and (iii) limited arbitrage capacity. The joint investor-motive pattern is most consistent with this family.

10 Conclusion

The paper’s main conclusion is that anomaly-return variance is primarily driven by **fundamentals-based trading (demand for characteristics)**, with a sizable secondary role for **latent (non-fundamental) demand**. Both channels are **disproportionately transmitted by small investors**. Institution-led mechanisms—**flows** and **industry rotation**—exist and matter in certain anomalies (especially issuance/size for flows and value/tangibility for industry), but they are smaller on average. The CF–DR integration further emphasizes that anomaly variation is **mostly cash-flow-news driven**, with small investors accounting for a large fraction of that CF component.

11 Conclusion

11.1 Summary

This paper concludes that **anomaly-return variation is primarily a demand-side phenomenon driven by (i) changes in fundamentals-based demand for characteristics and (ii) an additional non-fundamental “latent demand” layer**, and that **the investor types transmitting these two forces are not symmetric**.

The empirical narrative is a consistent chain:

1. **Build a unified demand-based decomposition.** The authors estimate a two-tier nested-logit demand system (industry $\rightarrow$ stock) using holdings data, recover demand elasticities, and then solve for equilibrium prices. They perform counterfactual price updates that change *one block at a time* (characteristics demand, latent demand, AUM/flows, and industry rotation) and map the resulting price changes into anomaly-portfolio return components.
2. **Rank trading motives on a common scale.** Averaged across 47 anomalies (Table 1), the variance shares are:

$$\underbrace{40\%}_{\text{demand for characteristics}} + \underbrace{19\%}_{\text{latent demand}} + \underbrace{13\%}_{\text{industry effects}} + \underbrace{3\%}_{\text{flow-induced trading}} \approx 75\% \text{ of total anomaly variance.}$$

Hence, the dominant forces are *characteristics demand* and *latent demand*; flows are small on average.

3. **Show that different anomalies load on different motives.** At the category level (Fig. 1), characteristics demand is especially large for momentum (about 64%) and remains high for profitability, size, and issuance, while latent demand becomes large precisely where characteristics demand is smaller (notably value and investment). Flow-induced trading becomes material mainly in issuance and size, and industry rotation peaks in tangibility and value.
4. **Identify which investors transmit which motives.** The investor-by-motive decomposition shows a sharp pattern:
 - **Characteristics demand is primarily transmitted by “small investors”** (a refined residual: non-13F after removing blockholders and short positions), with much smaller average contributions from large institutions.
 - **Latent demand also tilts toward small investors** in the categories where it matters most, with mutual funds as an important secondary contributor.
 - **Flow-induced trading is mostly institutional** (mutual funds and investment advisors).
 - **Industry rotation is institution-led**, with mutual funds, advisors, and banks accounting for most of the industry component.
5. **Connect the demand decomposition to a CF–DR news lens.** The paper finds that **cash-flow news dominates discount-rate news** in anomaly innovations: CF news explains about 58%–71% of anomaly-variance across categories (average $\approx 62\%$). Moreover, **small investors account for about 26% of CF-related variance** in anomaly returns (about 14% through characteristics demand and 12% through latent demand), which is nearly half of the total CF component.

11.2 Key quantitative takeaways (numbers worth memorizing)

- **Motive ranking (Table 1, mean across all anomalies):** demand for characteristics 40%, latent demand 19%, industry effects 13%, flows 3% (these four sum to $\approx 75\%$).
- **Category highlights (Fig. 1):** characteristics demand is largest for momentum (about 64%); latent demand is large in value (about 32%) and investment (about 33%); flows become material mainly in issuance and size (about 8% each); industry effects peak in tangibility (about 19%) and value (about 16%).
- **Who transmits the fundamentals channel (Table 2, mean across all anomalies):** small investors account for about 27% of total anomaly variance through characteristics demand; major institutions (banks and mutual funds) are each about 5% on average through this channel.
- **Who transmits the flows channel (Table 4, mean across all anomalies):** investment advisors contribute about 2% and mutual funds about 3% on average; small investors are near zero (slightly negative).
- **CF–DR conclusion:** cash-flow news accounts for about 58%–71% of anomaly variance across categories (average $\approx 62\%$), and small investors explain about 26% of the CF-related part (roughly 14% characteristics CF + 12% latent CF).

11.3 Why these channels appear in the decomposition

The intuition is that the demand system separates *what investors want* from *how they reallocate*:

1) Demand for characteristics as fundamentals-based trading. In a demand-based asset-pricing view, coefficients on stock characteristics summarize how investors trade off price against observable attributes. Time-variation in these coefficients reflects changes in investors’ *beliefs* or *pricing kernels* related to cash-flow growth and/or risk exposures. Hence, when anomaly returns covary strongly with the “characteristics demand” counterfactual, the anomaly is largely a **fundamentals-based demand channel**.

2) Latent demand as non-fundamental shifts in tastes or constraints. Latent demand is the residual component of demand not explained by observed characteristics. Economically, it can represent attention/salience, sentiment, disagreement, speculative tastes, or trading constraints that shift demand without a corresponding change in fundamentals proxies. Latent demand becomes large exactly where characteristics demand is smaller (e.g., value and investment categories), consistent with a **second layer beyond observables**.

3) Why “small investors” matter for the first two channels. The paper’s investor classification isolates a residual long-only group after removing blockholders and short sellers from non-13F ownership. While this residual may still contain some non-13F institutions, its behavior matches canonical retail-style patterns in the data: it is the main transmitter of characteristics-driven demand and, in select categories, latent demand. A natural interpretation is that this group has stronger exposure to attention/sentiment and less sophisticated processing of signals, so its reallocations move prices in anomaly-salient directions.

4) **Why flows and industry rotation look institutional.** Flow-induced trading is mechanically linked to AUM changes, which is a dominant margin for delegated institutional vehicles (mutual funds and investment advisors). Industry rotation (outer-nest shifts) is also consistent with institution-level allocation decisions (sector mandates, benchmarking, style rotation), so it is natural that these motives are institution-led even if they are not the first-order driver of anomaly variance on average.

11.4 How the conclusions map to anomaly theories

The decomposition speaks to three broad theory families:

- **Flow-based theories:** predict a large flow-induced component and institutional dominance. The evidence finds flows matter in specific categories (issuance and size) but are *small on average*.
- **Risk-based theories:** can be consistent with large characteristics-demand shares if characteristics proxy for priced risk and/or cash-flow growth. The CF–DR split suggests both cash-flow and discount-rate channels are quantitatively relevant within the fundamentals-based component.
- **Heterogeneous-investor / mispricing theories:** predict a meaningful role for non-fundamental demand and a prominent role for less sophisticated investors. The joint finding that (i) latent demand is sizeable and (ii) small investors transmit a large part of both the characteristics and latent components aligns naturally with heterogeneous-investor mechanisms.

11.5 Takeaway

Li, Sokolinski, Tamoni (2026) show that anomaly-return variance is dominated by changes in demand for characteristics (about 40%) and latent demand (about 19%), with smaller roles for industry rotation (about 13%) and flows (about 3%); the fundamentals and latent channels are disproportionately transmitted by small investors, flows are mostly institutional, and anomaly innovations are primarily cash-flow-news driven (average $\approx 62\%$ CF share), with small investors explaining a large fraction of that CF component.